



Monitoring of seawater immersion degradation in glass fibre reinforced polymer composites using quantum dots



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ABSTRACT

In this paper, a simple and innovative method is proposed to monitor the effects of seawater ageing in glass fibre reinforced polymer (GFRP) composites using quantum dots (QDs). The cadmium sulfide (CdS) QDs modified with 3-(trimethoxysilyl)-1-propanethiol (MPS) were firstly synthesized, and then the modified CdS QDs were labeled on the glass fibre by chemical bond. GFRP composites with QDs at the interface between the fibre and matrix were obtained. The single-fibre-fragment tests of specimens were done to verify the modified CdS QDs did not affect the interfacial bonding of the matrix and fiber. The effects of seawater ageing on the adhesion behaviour between the fibre and the matrix of QD/GFRP hybrids were investigated based on confocal fluorescence microscopy images. After 6 months of immersion in seawater, the glass transition temperature (T_g), tensile strength, and flexural strength of specimens decreased 2.5%, 13.8%, and 9.8%, respectively. Furthermore, the thermal and mechanical properties of specimens can be calculated by the fluorescence intensity quantitatively.

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1. Introduction

Fibre reinforced polymer (FRP) composites have attracted great attention in marine applications, such as yacht marina, offshore oil and gas platforms, rapid island buildings, and so on [1–4]. They are suitable materials for structural applications where high performance and long service life are required. FRP composites used in marine applications should retain their mechanical properties. However, FRP composites are susceptible to heat, moisture, NaCl molecules, algae, and ultraviolet (UV) radiation when they are exposed to the marine environment. The major issue of FRP composites used in the marine environment is that the interface between the fiber and the matrix would be debonded, thus the thermo-mechanical properties of composites are affected [5–10].

To date, a great deal of investigations have been concentrated on the effects of seawater exposure on the durability of FRP composites. Bian et al. [11] investigated the water absorption ability and

mechanical properties of GFRP specimens immersed into artificial seawater. After 42 days immersion, the tensile strength, the flexural strength, and the ILSS of composite specimens decreased 13%, 43%, and 50%, respectively. Similar observations were made by Hammami and Al-Ghuilani [12] in which 19.8% and 8.3% declines occurred in flexural strength and modulus of glass/vinylester composites after 6 months exposure in seawater at room temperature. The research of Aldajah et al. [13] indicated that the reductions of flexure stiffness of glass/polyester composites exposed to tap water and sea water environments during 2000 h were 29.4% and 28%, respectively. Springer et al. [14] recorded that the tensile strength of glass/vinyl ester composites decreased by 6% and 19% exposed to saturated NaCl solution at 23 °C and 93 °C during 6 months, respectively. In a 10-month study, Visco et al. [15] found that the stiffness and strength of vinyl ester resin based composites were higher than those of isophthalic polyester resin based composites. The flexural strength and flexural modulus decreased by 22% and 13%, respectively, for isophthalic polyester resin based composites. While, for vinyl ester resin based composites, the flexural strength was unchanged, and the flexural modulus increased by 20% in the same condition. In the research of Abdel-Magid et al. [16] on glass/polyurethane composites, the specimens exposed at 65 °C showed higher loss of tensile strength

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(~31%) in seawater for 1 year compared to that of the specimens exposed at room temperature (~19%). Xu et al. [17] found that the compression-after-impact strengths of E-glass/vinyl ester composites reduced by around 10% over 29 months seawater exposure. Additionally, Xian et al. [18] investigated the effects of sustained bending strain coupled with seawater immersion on the mechanical property of CFRP plates. The results indicated that the tensile strength for specimens in the highly strain (50% strain level) decreased by 18% and 13% after 2 weeks of immersion in water and seawater, respectively. In general, mechanical properties of FRP composites in the marine environment were compromised. Liao et al. [19] highlighted that the interface between the fiber and matrix played an important role in controlling FRP performance in immersion environment. The performance degradation is a consequence of chemical reaction and plasticization when water penetrates the interface [20–25]. Any microscopic damages such as fibre fractures or fibre-matrix debonding can lead to stress concentrations when FRP composites are exposed to the marine environment. Hence, it is necessary to gain insight into the relationship between the microscopic damages and the mechanical properties of FRP composites under the marine environment.

Up to now, many approaches for detect barely visible damages have been achieved by using X-rays [26], ultrasound [27], Raman spectroscopy [28], atomic force microscopy [29], bragg grating (FBG) sensor [30,31], fluorescent protein sensor [32,33], and so on [34,35]. Among of them, fluorescent protein, which is a kind of the fluorescent labelling agent, has been successfully applied to report mechanical damages in FRP composites owing to its high biocompatibility and good fluorescent stability. However, as a practical point of view, fluorescent proteins are difficult to produce in large quantities because most of them are extracted from the living organisms. On the other hand, quantum dot (QD), which is another kind of the fluorescent labelling agent, has been applied in biological labels due to its unique fluorescence labelling feature and mass-production suitability [36,37]. Recently, QD based composites have exhibited ultrasensitive response to Pb^{2+} , Cd^{2+} , Zn^{2+} and Mn^{2+} ions. Hence an available route in ultrasensitive ion probes and environmental detection was provided [38]. In addition, carbon nanotube (CNT) has been applied in humidity sensors due to its high electrical conductivity and sensitivity to water vapor doping [39,40]. Although much literature has been reported on the damage detection of FRP composites, unfortunately, there is little information on a facile method to detect the damage arising from seawater ageing of FRP composites by the use of QDs.

In this study, a simple and efficient strategy for sensing seawater immersion damages in FRP composites is reported. QDs were used as damage-sensitive links between glass fibre and isophthalic polyester resin in FRP composites (Fig. 1). Separation of isophthalic polyester resin from glass fibre would result in unfolding of QD when subjected to seawater environment, leading to the loss of fluorescence. To synthesize QD/FRP hybrids, QDs were modified with 3-(trimethoxysilyl)-1-propanethiol (MPS), thus QDs were immobilized on the glass fibre by the reaction between the methoxy group of MPS and the hydroxyl group of the glass fibre. The functionalized QDs-coated glass fibre was then embedded in the isophthalic polyester resin. Perhaps more importantly, a novel fluorescent probe for seawater ageing damages based on the changes of fluorescence intensity of QD/FRP hybrids was developed.

2. Materials and methods

2.1. Materials

Isophthalic polyester resin (HS-2100) was supplied by Changzhou Huake Resin Co. Ltd. (China). Bidirectional ($0^\circ/90^\circ$) E-glass

plain woven fabrics (500 g/m²) were collected from Sinoma Nanjing Fiberglass Research & Design Institute (China). Cadmium chloride ($CdCl_2 \cdot 2.5H_2O$), 3-(trimethoxysilyl)-1-propanethiol (MPS), sodium sulfide ($Na_2S \cdot 9H_2O$), and *N,N'*-dimethylformamide (DMF) were purchased from Aldrich. All chemicals were used as received, without further purification.

2.2. Synthesis of CdS QDs-labeled glass fibre fabric

The cadmium sulfide (CdS) QDs modified with MPS were synthesized in DMF based on our previous work [37]. Firstly, 1.370 g (6 mmol) $CdCl_2 \cdot 2.5H_2O$ was dissolved in 2.0 mL deionized water. When the cadmium salt was completely dissolved, the cadmium chloride solution was mixed with 3.534 g (18 mmol) MPS in 120 mL DMF and stirred vigorously for 10 min. Then, 0.432 g (1.8 mmol) $Na_2S \cdot 9H_2O$ dissolved in 1.0 mL deionized water was slowly added dropwise into the above solution with stirring. Hence, the color of the above solution turned yellow immediately, and the reaction was carried out for an additional 4 h under room temperature. To purify the as-synthesized CdS QDs, the resulting solution was dehydrated by distillation under high vacuum, along with the addition of fresh DMF for several times to remove the insoluble NaCl. The E-glass fibre fabrics were immersed in CdS QDs solution ($Cd^{2+} = 0.5$ M) for a period of 5 h. Finally, the samples were dried in a vacuum oven at $90^\circ C$ until the weight of them was constant.

2.3. Composite processing

Composite laminates, consisting of CdS QDs-labeled E-glass plain woven fabrics with isophthalic polyester resin, were made by manual lay-up process. Methyl ethyl ketone peroxide (MEKP, 1 wt%) and cobalt naphthenate (0.1 wt%) were added as a curing agent and an accelerator, respectively. All the specimens were cured for 24 h under room temperature and post-cured for 20 h at $80^\circ C$. The ageing was performed in the artificial seawater (CNSIC Marine Biotechnology Co., Ltd, China) or deionized water under room temperature for 6 months.

2.4. Interfacial property test

To verify whether the modified CdS QDs will affect the interfacial bonding of the matrix and fiber, the single-fibre-fragment tests (SFFT) of the composites with and without modified CdS QDs were done. The dump-bell shaped SFFT specimens were prepared according to Ref. [41]. A single fibre with or without modified CdS QDs was embedded in the middle of the high temperature resistant silicone mold with a metal weight 10 g at each end of the fibre to pre-strain the fibre. Then the specimen was filled with resin. After curing at room temperature for 6 h, move the specimens into an oven with $80^\circ C$ for 1 h. And the SFFT test was tensioned with the cross-speed of 1 mm/min using the Zwick/Roell universal testing machine, with a load cell of 20 kN capacity. This experiment was done under a microscope so that the fragmentation process can be observed in-situ. The interfacial shear strength (IFSS) was calculated according to Eq. (1):

$$\tau = \frac{\sigma_f}{2} \left(\frac{d}{l_c} \right) \quad (1)$$

where σ_f is the fibre strength at the critical length, d is the fibre diameter and l_c is the critical fragment length of fibre. The “critical fibre length” is defined as the shortest fragment length that breaks due to a stress application. Eq. (2) was used to determine the critical length of fibre:

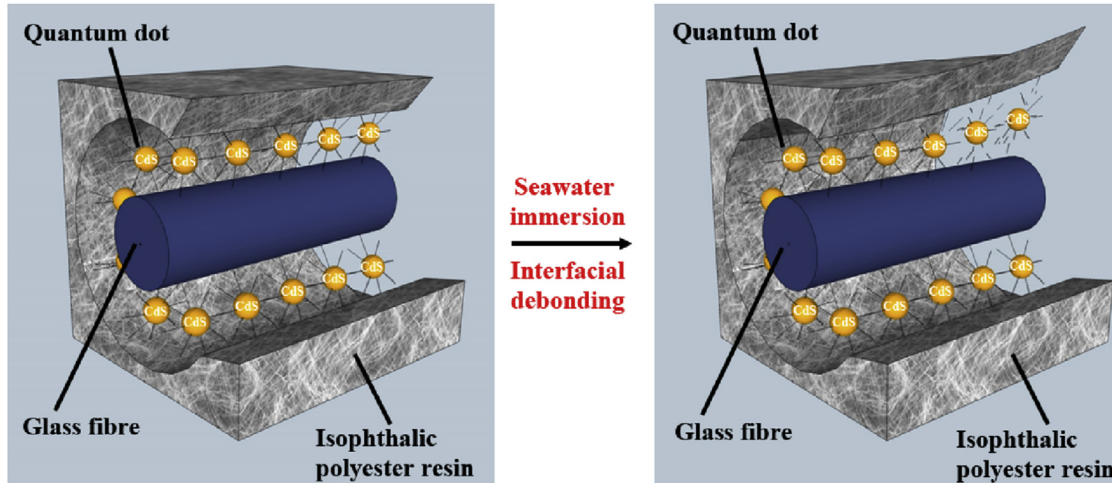


Fig. 1. Schematic detection procedure of fibre-matrix debonding in glass fibre reinforced composites under seawater immersion environment by quantum dots. (Schematic not to scale).

$$l_c = \frac{4}{3} \times \bar{L} \quad (2)$$

where $\bar{L}$ is the average fragment length.

2.5. Water absorption test

Water absorption measurements were performed by gravimetric analysis according to Ref. [20]. At regular time intervals, the specimens were taken out, wiped of excessive water with filter paper, weighed, and returned to the swelling medium. The test was continued until the constant weight of a sample was achieved. The moisture absorption M_t at any time can be calculated by Eq. (3) [42].

$$M_t \left[\frac{m_t - m_i}{m_i} \right] \times 100 \quad (3)$$

where m_i is the initial mass of the specimen prior to immersion, and m_t is the mass of the specimen at immersion time t .

2.6. Glass transition temperature test

Differential scanning calorimetry (DSC) test was performed on the specimens to determine the glass transition temperature (T_g) during ageing in seawater. A DSC 200 F3 Maia (NETZSCH) was used. The specimens were heated from room temperature to 250 °C, with a rate of 5 °C/min under nitrogen flow (100 ml/min). In all the measurements, the T_g s were taken from the second scans.

2.7. Fluorescence properties test

The fluorescence properties of specimens were determined with ultraviolet–visible (UV–vis) spectra (Perkin-Elmer Lambda 900 UV–vis spectrometer) and photoluminescence (PL) spectra. PL spectra of specimens were measured with a Varian Cary Eclipse fluorescence spectrophotometer under room temperature. Confocal microscopy images were taken with the Leica SP5 fluorescence lifetime imaging microscopy (FLIM) system.

2.8. Scanning electron microscope (SEM)

Structures of specimens before and after ageing were investigated by scanning electron microscope (SEM) with a QUANTA 200

(Philips-FEI, Holland). The specimens included unconditioned and conditioned samples which were immersed in seawater for 2, 4, and 6 months. All specimens were cut and coated with a gold film to improve their electrical conductivity before being observed by SEM.

2.9. Mechanical test

In order to study the mechanical performance of specimens, the tensile and flexural tests were conducted. After each exposure period, five specimens were tested for each condition. Tensile tests were conducted according to ASTM D3039/D3039M - 14 [43]. A universal testing machine with 50 kN capacity was used. The flexural tests were performed in a universal testing machine according to ASTM D7264/D7264M - 15 [44]. Before applying the load, the appropriate load equipment was installed.

3. Results and discussion

3.1. Interfacial property of composites

The values of IFSS test of composites with and without modified CdS QDs are shown in Table 1. For specimen with and without modified CdS QDs, the characteristics showed the same values. According to Eq. (1) and Eq. (2), the calculated interfacial bond strength for both specimen with and without modified CdS QDs was similar. The results indicate that the modified CdS QDs could not affect the interfacial bonding of the matrix and fiber, which is very critical in the mointing process of seawater immersion degradation in glass fibre reinforced polymer composites using modified CdS QDs as sensors.

3.2. Changes in glass transition temperature (T_g)

The measurement of T_g is necessary because it determines the critical service temperature of the composites. The specimens ageing in water and seawater at different immersion time were subjected to thermal analysis. For each specimen, two heating ramps ranging from room temperature to 250 °C were carried out. Here, the second scan was done to determine T_g . Fig. 2 shows the evolution of the T_g of specimens immersed in water and seawater environments, respectively. The T_g of the unaged specimen was 78.5 °C, whereas the T_g s of specimens after immersed in water and

Table 1
Characteristics of single-fibre-fragment test of composites with and without modified CdS QDs.

With or without modified CdS QDs	Strain F (N)	The number of fibre breaks	Max length $L_{\max}$ (μm)	Min length $L_{\min}$ (μm)	Average length L_{ave} (μm)
Without	250.24	9	1134	854.2	976.18
With	250.24	9	1134	854.2	976.18

seawater for 6 months were 76.2 °C and 76.5 °C, respectively. The decreases in T_g s are probably due to the effect of water plastization. The results can be ascribed to water penetration into the network structure of the resin, which would cause an increase in free volume. Hence a lowering of the T_g was achieved [45,46]. The decreases in T_g s reflect degradation of mechanical properties.

3.3. Moisture absorption

After 180 days ageing in water and seawater conditions, the experimental moisture absorption contents of GFRP composites as a function of the square root of immersion time are presented in Fig. 3. Each data point represents the average value obtained from five replicated specimens. The moisture absorption curve indicates a multistage process, which is dissimilar to the classic Fick's law. Initially, the moisture absorption of GFRP composites increases linearly with the square root of time, and then increases slowly followed by leveling off moisture absorption, which is well fitted to the classic Fick's law [47,48]. However, when immersion time increases, a dramatic mass loss can be found for the specimens immersed in both water and seawater, which deviates from Fick's law. The mass loss of GFRP composites is attributed to the leaching of un-reacted monomers and hydrolysis of ester bond [49,50]. At immersion time of 180 days, the moisture absorption of specimens immersed in water and seawater was 1.21 and 0.57, respectively. The results suggest that the equilibrium moisture absorptions in seawater are slightly lower than those in water, which is consistent with the results reported in Ref. [51].

3.4. Mechanical behaviour

The results obtained in the mechanical tests of GFRP composites, as a function of immersion time and ageing conditions, are illustrated in Figs. 4–7. The tensile strength and modulus of aged

specimens versus time of immersion in water and seawater are shown in Figs. 4 and 5. As shown in Fig. 4, for both ageing conditions, an overall rapid decrease in tensile strength is observed during the 6 months of immersion in water and seawater. After 6 months, GFRP composites had a tensile strength loss of 15.5% and 13.8% for water and seawater immersion conditions, respectively. In addition, it can be seen from Fig. 5 that the two curves show a slight fluctuation of tensile modulus of specimens for both ageing conditions. The tensile modulus reduced by 2.7% and 2.1% for specimens after 6 months ageing in water and seawater, respectively.

The three-point bending testing method has also been utilized

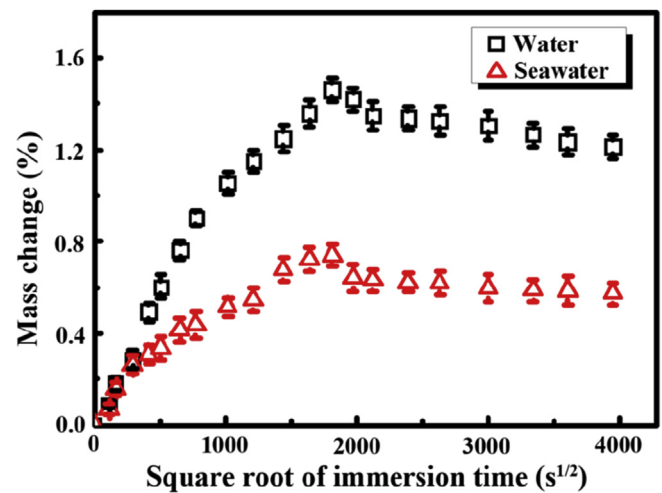


Fig. 3. Moisture absorption of GFRP composites as a function of square root of immersion time for water immersion and seawater immersion.

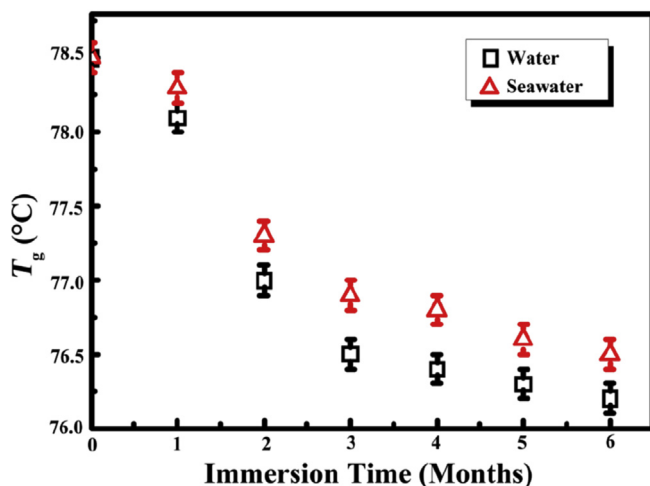


Fig. 2. Glass transition temperature (T_g) of GFRP composites as a function of immersion time for water immersion and seawater immersion.

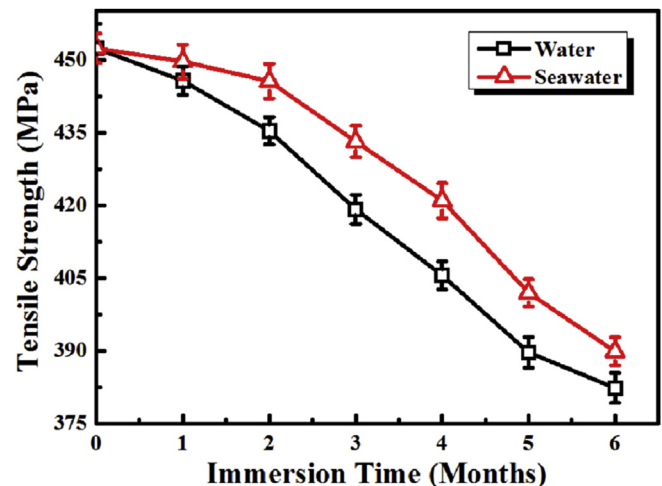


Fig. 4. Tensile strength of GFRP composites vs. immersion time for water immersion and seawater immersion.

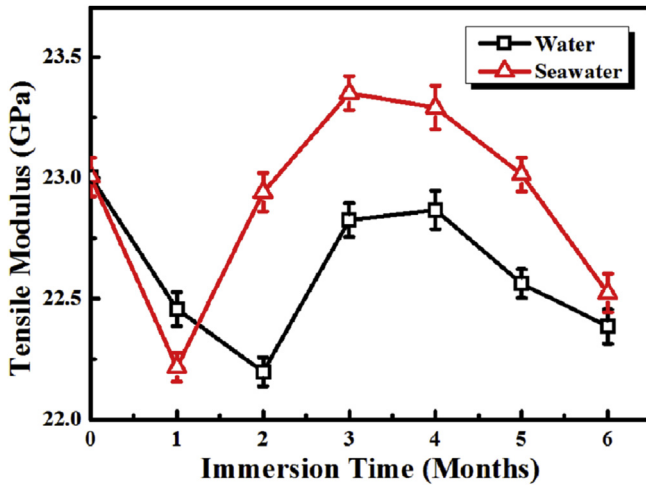


Fig. 5. Tensile modulus of GFRP composites vs. immersion time for water immersion and seawater immersion.

to analyze the degradation of GFRP composites in exposure conditions, because they can comprehensively reflect the mechanical properties of the composites [52]. Flexural tests of GFRP composites were performed at the age of 0, 1, 2, 3, 4, 5, and 6 months immersion. Figs. 6 and 7 show the changes in flexural strength and modulus of specimens exposed to water and seawater conditions. It can be seen from Figs. 6 and 7 that after 6 months, the loss of flexural strength was 13.5% and 9.8% for specimens immersed in water and seawater, respectively. Similarly, after 6 months, the loss of flexural modulus was 5.8% and 4.6% for specimens immersed in water and seawater, respectively.

It is obvious that the tensile and flexural strength of aged specimens after every period decline compared to virgin specimens. However, the changes in tensile and flexural modulus do not appear to follow a clear trend. Similar behaviour was observed by Feng et al. [53]. It is evident that long-term immersion of GFRP composites leads to a decrease in tensile and flexural strength. Additionally, the results indicate that the tensile and flexural strength of specimens aged in water are lower than those of specimens aged in seawater, which is consistent with the results in

Ref. [13]. The deteriorations in tensile and flexural properties can be attributed to water diffusion, resulting in degradation of resin, fibre fractures, and fiber-matrix debonding [54,55]. Higher moisture absorption in water is an important factor for the degradation in tensile and flexural properties. Moreover, the presence of NaCl molecules in seawater prevents the water molecules diffusion from the matrix to the fibres, which leads to less degradation in tensile and flexural properties of specimens exposed to seawater.

3.5. Morphology analysis

The adhesion behaviour between the fibre and the matrix plays an important role in the loading transfer from the matrix to the fibre. This aspect can be investigated by SEM measurement. A representative SEM micrograph of control specimen is shown in Fig. 8a. The microstructure of control specimen shows a strong adhesive between the fibre and the matrix. The significant fibre pull-out and uneven matrix fragments can not be found, which indicates good interfacial adhesion. The micrographs in Fig. 8b–d shows the surfaces of the aged specimens with 2, 4, and 6 months, respectively. As we expect, the fibre breakage as well as fibre pull-out and a higher degree of polymeric deterioration occurs due to the increase in immersion time. For specimens with immersion of 6 months, Fig. 8d shows that debonding occurs between the fiber and matrix and consequent fiber pull-out. In the case of moisture, it is obvious that resin hydrolysis and debonding at the interface are the major reasons for the degradation of flexural properties for GFRP composites.

3.6. Damage self-reporting of aged composites

According to the unique fluorescence labelling feature of QDs, QDs embedded at the interface between the fiber and matrix play an important role in sensing the damage process of the aged GFRP composites. In order to investigate whether QDs are able to evaluate microscale damages within composite materials exposed to seawater immersion, specimens after seawater immersion under different immersion time were investigated by confocal fluorescence microscopy images. Damaged areas are usually hard to identify in the non-fluorescence images, especially in the case of the fabric. In contrast, the fluorescence channel reveals fractured fibres through an easily observable change in fluorescence patterns.

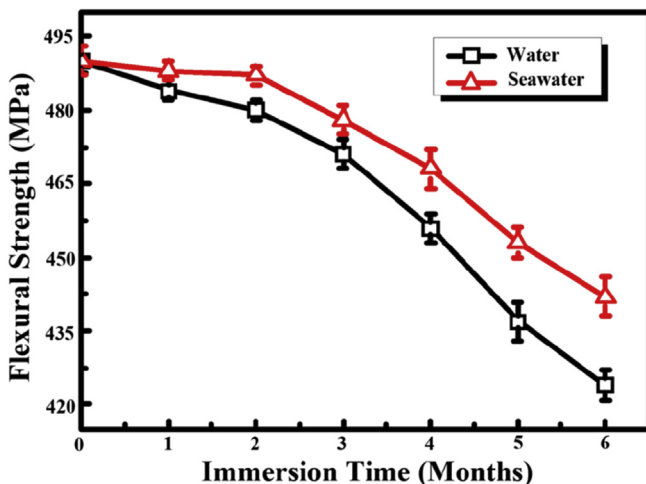


Fig. 6. Flexural strength of GFRP composites vs. immersion time for water immersion and seawater immersion.

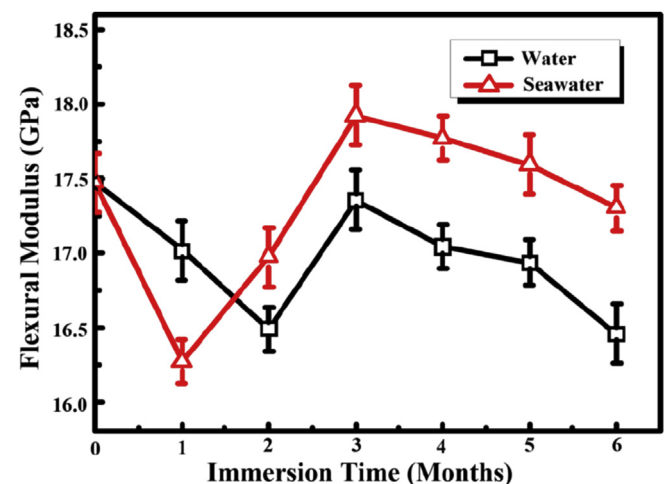


Fig. 7. Flexural modulus of GFRP composites vs. immersion time for water immersion and seawater immersion.

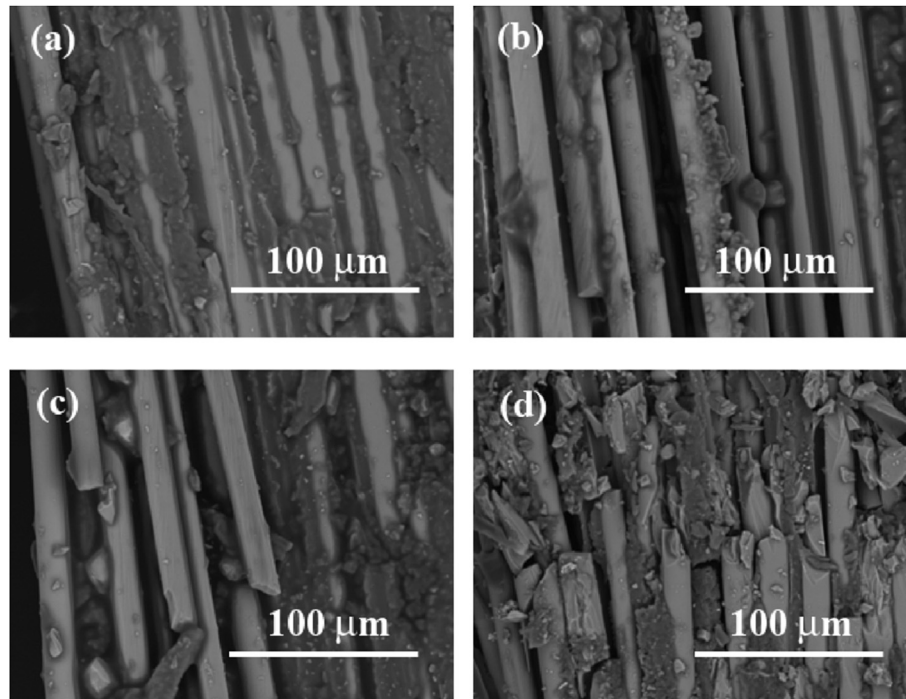


Fig. 8. SEM micrographs of GFRP composites surfaces at different immersion time: (a) 0 day, (b) 2 months, (c) 4 months, and (d) 6 months for seawater immersion.

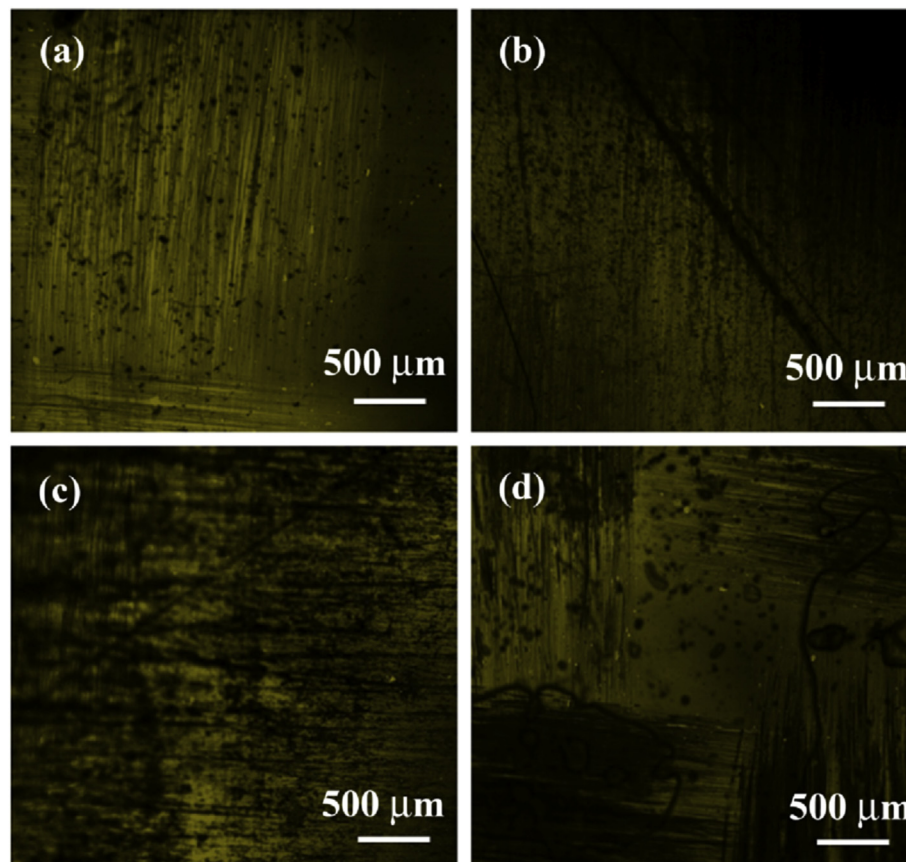


Fig. 9. Confocal fluorescence microscopy images of GFRP composites at different immersion time: (a) 0 day, (b) 2 months, (c) 4 months, and (d) 6 months for seawater immersion.

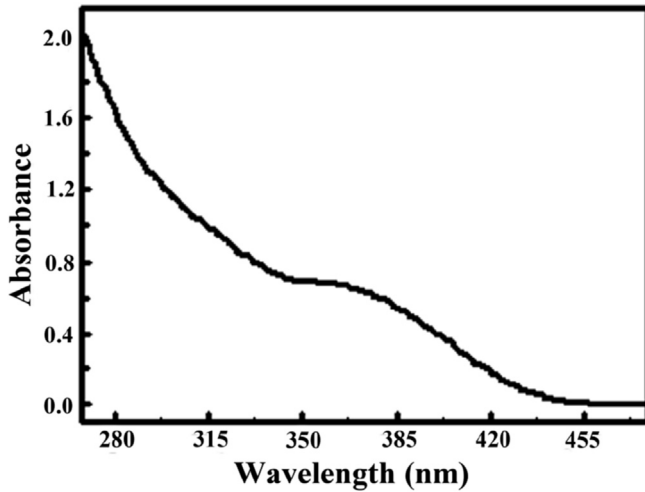


Fig. 10. UV–vis absorption of MPS- capped CdS QDs.

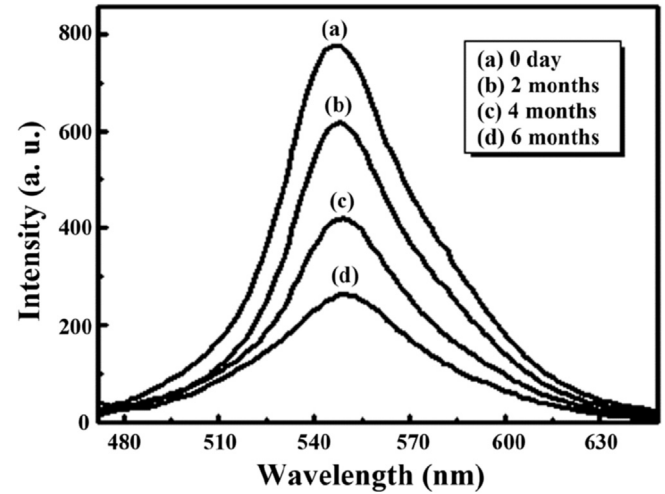


Fig. 12. PL spectra of GFRP composites at different immersion time: (a) 0 day, (b) 2 months, (c) 4 months, and (d) 6 months for seawater immersion.

As shown in Fig. 9a, the fluorescence patterns of unaged specimens indicate a good bonding behaviour between the fibre and the matrix. When the specimens expose to water immersion, the fluorescence disappears at the fracture site of fibre (Fig. 9b–d). Moreover, the vanished area of fluorescence of fibre becomes larger with the increase in the immersion time, which proves that the bonding property of interface is reduced at long-term immersion. The absence of fluorescence near fracture sites is due to the luminescence quenching phenomenon of QDs in seawater, which reflects the debonding behaviour of composites immersed in seawater for long-term. Therefore, QDs are able to detect zones of interfacial debonding immersed in water by loss in fluorescence.

The changes of PL intensity can be used as a quantitative index to report the damage of aged composites in addition to the confocal fluorescence microscopy images as a qualitative index. The UV–vis absorption and PL spectra of MPS-capped CdS QDs ($\text{Cd}^{2+} = 0.5 \text{ M}$, $\text{Cd}^{2+}/\text{MPS}/\text{S}^{2-} = 1:3:0.3 \text{ mol/mol/mol}$) are shown in Figs. 10 and 11. The absorption characteristic peak for the MPS-capped CdS QDs appears at 375 nm (3.31 eV). According to Brus's formula, the

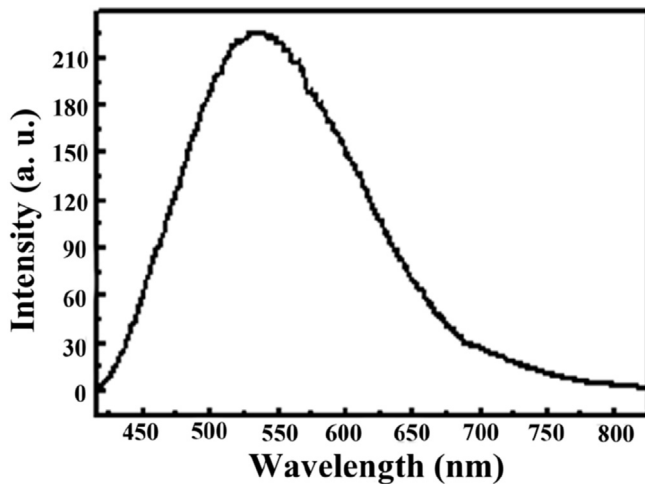


Fig. 11. PL spectrum of MPS-capped CdS QDs.

particle size of MPS-capped CdS QDs can be calculated by the wavelength of the maximum absorption peak in UV–vis spectra. As a result, the diameter of the MPS-capped CdS QDs is equal to 3.67 nm, which corresponds to the standard of QDs. The PL spectrum of MPS-capped CdS QDs shows a broad peak centered at 530 nm, mainly originating from the broad emission of the recombination of surface electrons and holes. The PL spectra of the QD/GFRP hybrids at different immersion time are displayed in Fig. 12. Interestingly, a red-shift of the PL peak varied from 530 nm for QDs to 550 nm for QD/GFRP hybrids can be observed. Initially, the PL intensity of specimen was 777 a. u. When the specimens were immersed in water after 2, 4, and 6 months, respectively, the PL intensity decreased to the value of 618, 422, and 265 a. u., respectively. The results indicate that the degree of the luminescence quenching is enhanced with the increase in the immersion time, which is consistent with the results of confocal fluorescence microscopy images (Fig. 9).

In order to investigate the relationship between the changes of PL intensity and the thermal and mechanical properties of QD/GFRP hybrids, Fig. 13 shows the effect of the relative fluorescence intensity (I/I_0 , where I and I_0 are the PL intensity of the aged specimen and the virgin one) on the relative glass transition temperature (T_g/T_{g0} , where T_g and T_{g0} are the glass transition temperature of the aged specimen and the virgin one), the relative tensile strength (H_t/H_{t0} , where H_t and H_{t0} are the tensile strength of the aged specimen and the virgin one), and the relative flexural strength (H_f/H_{f0} , where H_f and H_{f0} are the flexural strength of the aged specimen and the virgin one) of QD/GFRP hybrids for seawater immersion. As shown in Fig. 13a, the T_g/T_{g0} of the specimen varies from 1 at the data of I/I_0 of 1, rapidly down to 0.98 at the value of I/I_0 of 0.8, followed by a slight decline to 0.97 at the data of I/I_0 of 0.34. Meanwhile, a moderate downward trend of H_t/H_{t0} from 1 at the value of I/I_0 of 1 to 0.97 at the value of I/I_0 of 0.55 is shown in Fig. 13b, and then the data remains stable at approximately 0.97. Additionally, the change trend of H_f/H_{f0} is similar to the H_t/H_{t0} . The H_f/H_{f0} of the specimen varies from 1 at the value of I/I_0 of 1, markedly down to 0.87 at the figure of I/I_0 of 0.34. The results indicate that the fluorescence intensity can reflect the values of T_g , tensile and flexural strengths. Therefore, it is noted that the thermal and mechanical properties of the composites can be calculated by the fluorescence intensity quantitatively.

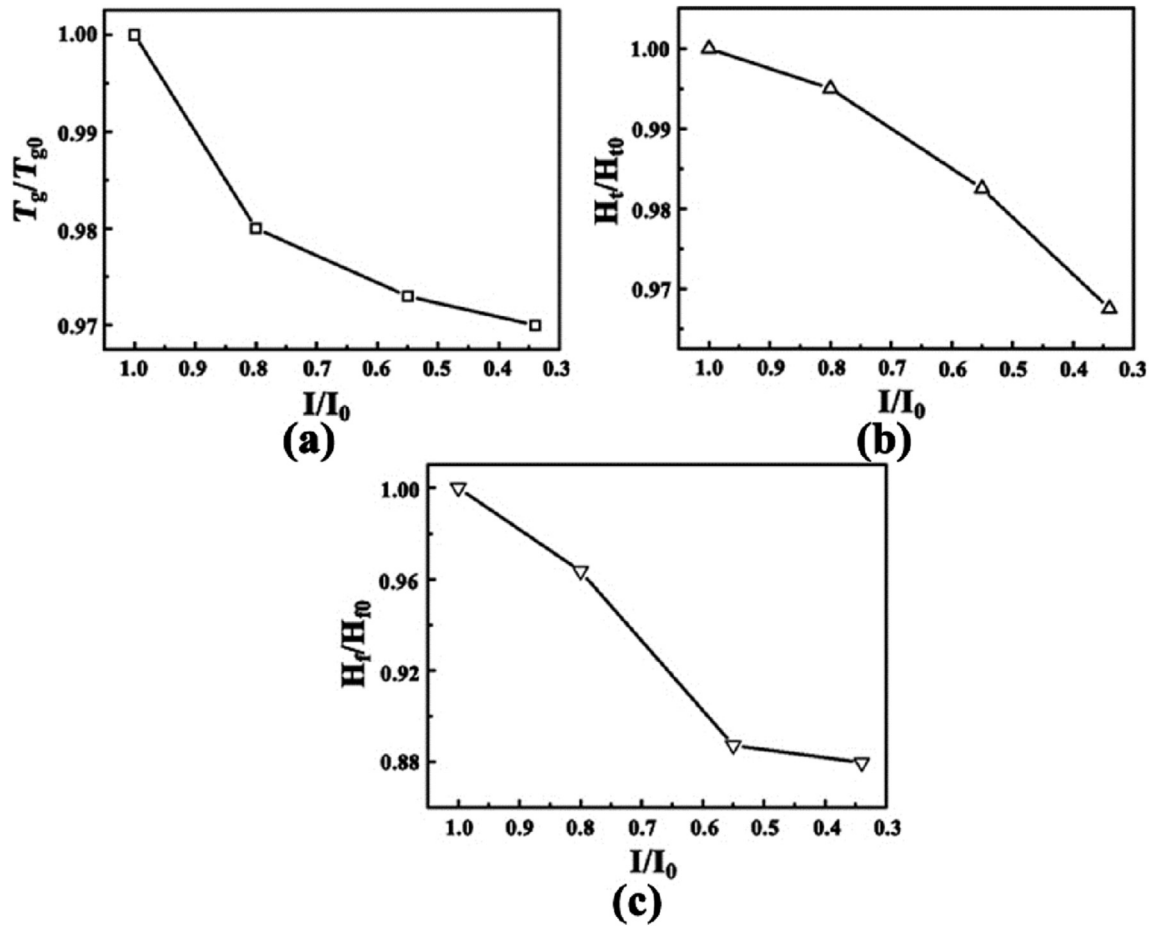


Fig. 13. Effect of the relative fluorescence intensity (I/I_0) on (a) the relative glass transition temperature (T_g/T_{g0}), (b) the relative tensile strength (H_t/H_{t0}), and (c) the relative flexural strength (H_f/H_{f0}) of GFRP composites for seawater immersion.

4. Conclusions

This paper demonstrates a simple strategy for the successful implementation of QDs as seawater immersion damage sensors in FRP composites. The main findings of this study are summarized as follows:

- (1) The moisture absorption increased initially and then decreased with immersion time because of hydrolysis of the ester groups of isophthalic polyester resins. It is shown that the initial water uptake capacity of all specimens has been demonstrated to follow the classic Fick's law, whereas a dramatic mass loss occurred with long time immersion due to severe hydrolysis of resin. Additionally, the equilibrium moisture absorption of specimens in seawater was slightly lower than that of specimens immersed in water.
- (2) During the stage of ageing in water and seawater conditions, GFRP composites present signs of plasticization, with a slight decline in T_g . After 6 months of ageing time, the T_g of specimens after immersed in water and seawater decreased 2.9% and 2.5%, respectively.
- (3) Immersion in water and seawater significantly affected the mechanical properties of GFRP composites. After 6 months immersion in water and seawater, the tensile strength of GFRP composite showed a significant downward trend by 15.5% and 13.8%, respectively. Whereas, the tensile modulus was less affected than tensile strength in seawater or water.

Meanwhile, the losses of flexural strength and modulus were 13.5% and 5.8% for specimens immersed in water for 6 months, respectively, whereas, the losses of flexural strength and modulus were 9.8% and 4.6% for specimens immersed in seawater for 6 months, respectively. It is noted that resin hydrolysis and debonding at the interface are the major reasons for the mechanical properties of GFRP composites degradation.

- (4) By placing the modified QDs at the interface between the fibre and the isophthalic polyester resin, QDs-labeled FRP composites were obtained, which could be used to evaluate micro-scale seawater immersion damages by loss of fluorescence in the damaged areas and the changes of fluorescence intensity. On the one hand, the changes in vanished area of fluorescence of the specimens due to the luminescence quenching phenomenon of QDs in water solution can reflect the debonding behaviour of fiber and matrix. On the other hand, the loss of T_g , tensile property, and flexural property can be calculated by the reduction of fluorescence intensity.
- (5) Overall, it has been demonstrated that QDs are able to detect barely visible seawater immersion damages. It is expected that QDs can be used as the damage-sensitive link between the fibre and resin in GFRP composites immersed in seawater. A parallel approach to that of this investigation constitutes a promising way to detect barely visible damages by using QDs with additional advantages of facile and

efficient way compared with other techniques. However, some challenges, such as too costly and time consuming, need to be overcome.

- (6) Future work will focus on protections of GFRP composites against heat, moisture, seawater molecular, algae and ultra-violet (UV) radiation when they are exposed to the marine environment.

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